



Self-adaptable anti-interference scheme based on light-path blocking in an hybrid WiFi-VLC network

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Abstract: In this paper, a self-adaptable anti-interference scheme in a hybrid wireless fidelity (WiFi) and visible light communication (VLC) network is firstly proposed by light-path blocking. By human behavior characteristics, a user-device position relationship model is constructed to determine users' orientations. By the model, a strategy of choosing access points (AP) is present. By the strategy, communicating APs can be self-adaptively selected to match them with users' orientations. In the scheme, interference signals can be effectively blocked through the user's body to ensure normal communication. Finally, the effectiveness of the scheme has been demonstrated with simulations. Also, the scheme has its comparative advantages of not only saving energy, increasing SINR and lifting the availability of the hybrid WiFi-VLC network but also having faster response speed of network, higher efficiency of user access, easier implementation and lower cost.

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1. Introduction

The hybrid wireless fidelity (WiFi) and visible light communication (VLC) network has advantages of WiFi and VLC. It is one of the important networking forms in future communication. Nowadays, people pay more and more attention to its researches [1,2]. In the hybrid network, VLC access points (APs) are deployed intensively to avoid VLC being affected by light-path blocking and dead angle of illumination [3]. But this will cause serious interference between VLC APs [4,5]. Now, how to solve the problem of interferences between VLC APs has become a research hotspot.

Researches of anti-interference between VLC APs can be divided into three categories: interference avoidance at the transmitter end, interference cancellation at the receiver end or anti-interference in the spatial channel [6]. To the transmitter end: Jinyoung An et al. utilized time-hopping in VLC to avoid the collision of multiple signals [7]. Kaixiong Zhou et al. present a dynamic soft frequency reuse (SFR) scheme to avoid interference between VLC APs and adapt to different densities of APs [8]. To the receiver end: Chen Chen et al. put forward an optimized angle diversity receiver for different LED layouts to cancel interferences [9]. Muhammad Asim Atta et al. proposed a polarization-based approach to eliminate un-polarized optical interference between VLC APs [10]. Ahmed Adnan Qidan et al. applied blind interference alignment (BIA) in hybrid WiFi-VLC network to mask interference signals [11]. To the spatial channel, its researches are just beginning. At present, reports show that only Jona Beysens et al. designed a user-in-the-loop mechanism of guiding users to rotate their bodies to resist interferences between VLC APs. In this design, firstly, device is directly connected to APs. Next, the communication quality is detected. Then, users are guided to change their positions in order to block interference signals [12,13].

Here, according to the human behavior characteristics, this paper will propose a self-adaptable anti-interference scheme in hybrid WiFi-VLC network by light-path blocking. This scheme is

applicable to the indoor scenarios where the user does not change position for a short time. It can not only effectively block interference signals through the user's body to ensure normal communication, but also save energy, increase SINR and lift availability of the network.

2. System structure

2.1. Network structure

Hybrid WiFi-VLC network consists of a WiFi AP and several VLC APs. They are connected to a backhaul network through optical fiber [14]. The topology of the hybrid network is shown in Fig. 1.

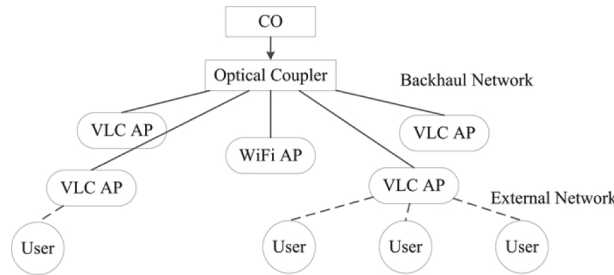


Fig. 1. The topology of the hybrid WiFi-VLC network.

2.2. VLC channel structure

The transmission channel of VLC is shown in Fig. 2 [15].

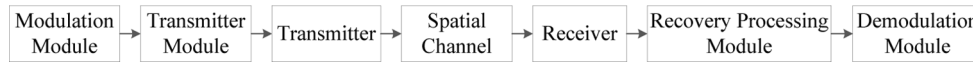


Fig. 2. VLC channel model.

In hybrid WiFi-VLC network, the spatial channel of VLC has two kinds: one is Line of Sight (LOS) link and the other is Not Line of Sight (NLOS) link. Due to the negligible influence of NLOS, only LOS is considered in this study. The spatial channel model is shown in Fig. 3 [16].

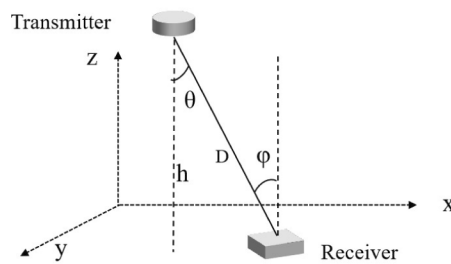


Fig. 3. Spatial channel model of LOS.

The receiver power of the line-of-sight link is computed by [16]:

$$p_r = p_t H(0) \quad (1)$$

Signal to interference plus noise ratio (SINR) is written by [16]:

$$SINR = \frac{\gamma^2 \sum p_r^2}{N_0 B + \gamma^2 \sum p_r'^2} \quad (2)$$

where, p_r is the received optical power from the communicating AP, p_t is the optical power of transmitter, $H(0)$ is the optical channel's total Direct Current (DC) attenuation, γ is the photo detector's responsivity, $N_0 B$ is noise, p_r' is the received optical power from the interfering AP.

3. User-device position relationship model

3.1. Behavior characteristics of human beings

In the indoor scenarios where the user does not change position for a short time. When users are moving, their orientations can be obtained by directions of their movements. When users are stationary, their orientations can't be obtained. In practice, their orientations are the directions from their coordinates to their mobile devices' coordinates. Mobile devices' coordinates can be obtained through feedback signals between mobile devices and VLC APs. But users' coordinates are unknown. In fact, there is a certain relationship between users' and devices' position distributions in real life.

When users use fixed devices, they tend to face the nearest fixed device. When no fixed device is in a scene, users' positions tend to distribute as a polygon due to the social characteristics of human beings. It means that users will face the middle of the polygon. But, if users' positions distribute as a straight line, users' orientations will change as the scene changes. Under this condition, users' orientations cannot be judged.

In the following, a position relationship model will be constructed to calculate users' coordinates and get their orientations.

3.2. Modeling

The modeling of the position relationship model can be divided into two stages. Stage 1 is to calculate users' coordinates when users face a fixed device. Stage 2 is to calculate users' activity-ranges when users form a gathering relationship with surrounding users.

Here, the vertex in the lower left corner of the room is set to the origin coordinates. For ease of description, define (x, y) , (x_f, y_f) and (x_u, y_u) as the mobile device's coordinate, the fixed device's coordinate and the user's coordinate respectively. When calculating one user's coordinate, *User* represents the user. *User1* and *User2* represent the closest user and the second closest user to *User*. *Device*, *Device1* and *Device2* represent *User*'s, *User1*'s and *User2*'s mobile devices respectively. Direction from *User*'s coordinate to his mobile device's coordinate is *User*'s orientation.

3.2.1. Stage 1: modeling under the condition that users face fixed devices

For ease of description, define $d_{f_{\max}}$, $d_{f_{\min}}$, R_f and k_f as the maximum threshold of distance, the distance between *Device* and the nearest fixed device of *User*, the relevance between *User* and the nearest fixed device and the slope of the line located by *Device* and the fixed device respectively.

To accurately calculate users' coordinates under the condition that users face fixed devices, four steps will be taken. These four steps are as follows:

Step 1: Here, $d_{f_{\min}}$ will be compared with $d_{f_{\max}}$ to judge whether the fixed device is in the range available to *User*. In practical, users have visual limitations and operating distance limitations. Therefore, there is the maximum threshold of distance ($d_{f_{\max}}$). If the distance between the user and the device is larger than $d_{f_{\max}}$, the user can't use the device. Here, $d_{f_{\max}}$ is set by the room area and the fixed device's screen size. When $d_{f_{\min}} \geq d_{f_{\max}}$, it means that the fixed device isn't in the range available to *User*. Then, Step2 will be started. When $d_{f_{\min}} < d_{f_{\max}}$, the next nearest fixed

device will be found. Then, Step 1 will be repeated. If no fixed device is in the range available to *User*, *User* will wait for next stage to judge his orientation.

Step 2: The relevance between *User* and the fixed device will be weighed to judge whether *User* is likely to use the fixed device. R_f is judged by the owner or the potential user of the fixed device. If $R_f = 1$, it means that the fixed device is relevant to *User*. That is, *User* may use the fixed device. Then, Step 3 will be started. If $R_f = 0$, the fixed device is irrelevant to *User*. So, the next closest fixed device will be found. Then, Step 1 will be repeated.

Step 3: The position relationship between *User* and the fixed device will be considered to judge whether the user is in front of the fixed device. When *User* is using a fixed device, he must be in front of the device. The position relationship between *User* and the fixed device can be judged by k_f . The judgment conditions are shown in Table 1.

Table 1. Judgment conditions

Orientation of fixed device	Condition	Orientation of fixed device	Condition
north	$x > x_f$	west	$x < x_f$
south	$y > y_f$	west	$x < x_f$
southeast	$\begin{cases} k_f > 1, & x \leq x_f \\ k_f < 1, & x > x_f \end{cases}$	northeast	$\begin{cases} k_f < -1, & x \leq x_f \\ k_f > -1, & x > x_f \end{cases}$
southwest	$\begin{cases} k_f > -1, & x \leq x_f \\ k_f < -1, & x > x_f \end{cases}$	northwest	$\begin{cases} k_f < 1, & x \leq x_f \\ k_f > 1, & x > x_f \end{cases}$

If the above conditions are all met, Step 4 will be started. Otherwise, the next nearest fixed device will be found and Step 1 will be repeated.

Step 4: The coordinate of *User* will be calculated. When the above steps are all finished, the coordinate of *User* can be calculated by:

$$\begin{cases} x_u = x + \frac{\text{span}(x-x_f)}{D} \\ y_u = y + \frac{\text{span}(x-y_f)}{D} \end{cases} \quad (3)$$

where D is the distance between *User* and the fixed device. span is the distance from *User*'s coordinate to his mobile device's coordinate. Normally, *User*'s orientation is the same as the direction from *User*'s coordinate to the fixed device's coordinate. So, *User*'s orientation can be gotten at the same time.

After the above four steps are all completed, this model will start a new cycle to calculate the next-user's coordinate. After the position relationship between all users' and all fixed devices has been gotten, the next stage will be started.

Obviously, the above indicates that users' coordinates and orientations can be gotten by the position relationship between users and fixed devices.

3.2.2. Stage 2: modeling under the condition that users face the crowd's center

When users not facing any fixed device get together, their positions distribute as a polygon. It means that users face the middle area of the polygon. In this situation, users' position will change in a small area. The coordinate of *Device*'s position is known. The range of *User*'s activity is an arc centered at *Device*'s position. The arc's radius is the distance between *User*'s coordinate and his mobile device's coordinate. On average, the radius is about 35cm. If the center, the radius, and the two endpoints of this arc are obtained, the function of the arc can be gotten. Here, just need to judge *User*'s orientation. So, only two endpoints' coordinates need to be calculated, the function of the arc doesn't need to be given. According to the coordinates of the two endpoints, the range of *User*'s activity can be obtained. And the actual coordinate of *User* is a point on the

arc. It means that *User's* orientation is roughly determined. When users face the crowd's center, the diagram of their position distributions is shown in Fig. 4.

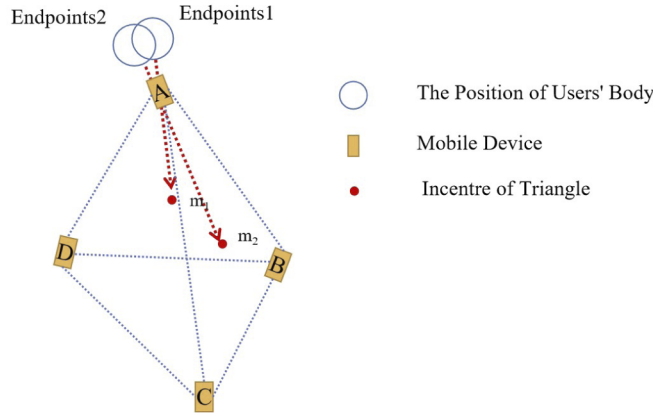


Fig. 4. The diagram of users' position distributions.

For ease of description, define (m_1, n_1) , (m_2, n_2) , (x_{u1}, y_{u1}) and (x_{u2}, y_{u2}) as the point m_1 's coordinate, the point m_2 's coordinate, the endpoint's coordinate obtained from the first calculation and the endpoint's coordinate obtained from the second calculation respectively. A, B, C and D represent users A, B, C and D's mobile devices respectively.

Take user A as an example, his activity-range can be gotten by two triangles' inner points. The method is described below.

A forms the triangle ABD with B and D. The point m_1 is the inner point of the triangle ABD. According to coordinates of m_1 and A, (x_{u1}, y_{u1}) can be calculated by:

$$\begin{cases} x_{ui} = x + \frac{\text{span}(x-m_i)}{D_{um}} \\ y_{ui} = y + \frac{\text{span}(y-n_i)}{D_{um}} \end{cases} \quad (4)$$

where D_{um} is the distance between the user's mobile device and the point m_i . span is the distance from (x_{u1}, y_{u1}) to the mobile device's coordinate. It is the first time to calculate the endpoint's coordinate. So, $i = 1$.

After obtaining (x_{u1}, y_{u1}) , A forms the triangle ABC with B and C. Similarly, (x_{u2}, y_{u2}) can be calculated by Formula 4. At this moment, $i = 2$. Accordingly, the activity-range of user A can be gotten.

The above indicates that Formula 4 can calculate users' activity-range when users face the center of the crowd. The construction of this model is discussed below.

When several users knowing each other get together, they are in the same user-group. The distance between any two of them is small. When users are in different user-groups, the distance between any two of them is large. If two user-groups are too close together, the calculation of the endpoint's coordinate is easily influenced by positions of users from the other group.

To make the calculation be more realistic, four steps will be taken. These four steps are as follows:

Step 1: Distances from *Device* to *Device1* and from *Device* to *Device2* will be respectively compared with the maximum threshold of distance and the user-related threshold of distance in order to judge whether users can form a user-group. Here, define $d_{\max}$, $d_{u\max}$, d_1 and d_2 as the maximum threshold of distance, the user-related threshold of distance, the distance between *Device* and *Device1* and the distance between *Device* and *Device2* respectively. $d_{\max}$ is set

according to normal social distance. $d_{u\max}$ is related to d_1 and d_2 . $d_{u\max}$ can be given by:

$$d_{u\max} = \delta_1 \times d_1 + \delta_2 \times d_2 + \frac{d_{\max}}{2} \quad (5)$$

where δ_1 , δ_2 and δ_3 are coefficients. δ_1 and δ_2 represent the weight of intimacy degree between *User* and *User1* and the weight of intimacy degree between *User* and *User2* respectively. $\delta_1 + \delta_2 = 1$.

If $d_1 < d_{\max}$, $d_2 < d_{\max}$, $d_1 < d_{u\max}$ and $d_2 < d_{u\max}$, these three users may be in the same user-group. Then, Step 2 will be started. Otherwise, *User* cannot form a user-group with anyone else. Then, this model will start to judge next *User*'s activity-range.

Step 2: If *User1* or *User2* of *User* faces a fixed device, distances from *User* to *User1* and from *User* to *User2* will be respectively compared with the minimum threshold of distance in order to judge whether they are in the same group. Here, define d_{uf} and $d_{uf\min}$ as the distance between *Device* and the mobile device of the user facing the fixed device and the minimum threshold of distance respectively.

When *User1* or *User2* of *User* faces a fixed device, *User* and *User1* or *User2* may not be in the same user-group. If $d_{uf} > d_{uf\min}$, the next *User1* and *User2* should be found. Then, Step 1 will be repeated. If $d_{uf} \leq d_{uf\min}$, following Step 3 will be started.

Step 3: The linearity of *User*'s, *User1*'s and *User2*'s position distributions will be calculated to judge whether the three users' positions are in a line. Here, define k_1 , k_2 , k_3 and *Line* as the slope of the line formed by positions of *Device* and *Device1*, the slope of the line formed by positions of *Device* and *Device2*, the slope of the line formed by positions of *Device1* and *Device2* and the linearity of *User*'s, *User1*'s and *User2*'s position distributions respectively. Closer values of k_1 , k_2 and k_3 are, closer *Device*, *Device1* and *Device2*'s position distributions are to the straight line. *Line* can be derived as follow:

$$Line = \frac{1}{3} \left(\frac{k_1}{k_2} + \frac{k_1}{k_3} + \frac{k_2}{k_3} \right) \quad (6)$$

If $Line \approx 1$, the next *User1* and *User2* will be found. Then, Step 1 will be repeated. Otherwise, following Step 4 will be started.

Step 4: The endpoint's coordinate will be calculated. When the above steps are all done, it means that *User* can form a triangle with *User1* and *User2*. So, based on the method described in Fig. 4, The endpoint's coordinate can be calculated by Formula 4.

If (x_{u1}, y_{u1}) has been calculated, this model will return to Step1 and start to calculate (x_{u2}, y_{u2}) . If only (x_{u1}, y_{u1}) can be calculated, it means that direction from (x_{u1}, y_{u1}) to (x, y) is the unique orientation of *User*. If (x_{u1}, y_{u1}) and (x_{u2}, y_{u2}) have been calculated, range of *User*'s activity can be obtained. Then, this model will start the next cycle to calculate the range of the next user's activity until the ranges of all users' activity can be gotten.

If neither stage 1 nor stage 2 can get *User*'s coordinate or ranges of activity, it means that *User*'s orientation can't be judged. This problem will be addressed in section 4.2.

Obviously, the above indicates that users' coordinates and orientations can be gotten by the position relationship between *User* and his surrounding users.

Now, the position relationship model has been constructed. The flow diagram of the model construction is shown in Fig. 5. Here, Fig. 5(a) is the flow diagram of the model construction when users face the fixed device, Fig. 5(b) is the flow diagram of the model construction when users face the center of the crowd.

From the above, it can be seen: Whether the user is facing the fixed device or the center of the crowd, his orientation can be determined by this model. After the positions and orientations of users has been determined, the network will go into the phase of selecting APs.

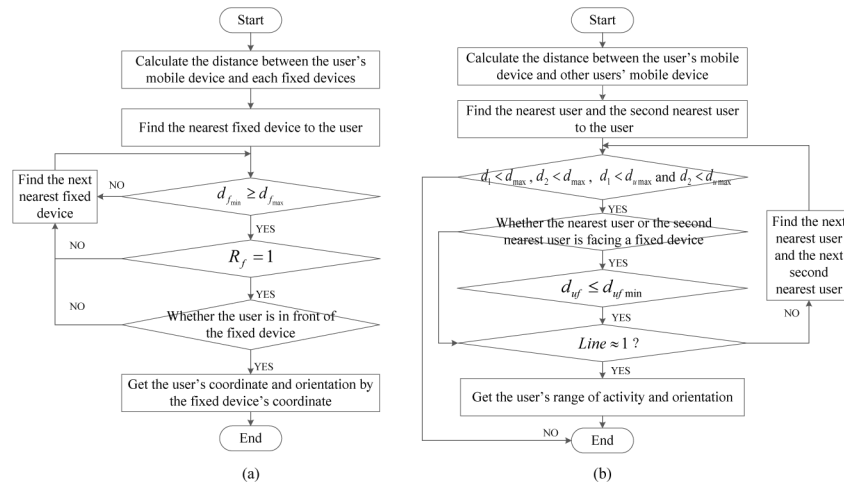


Fig. 5. The flow diagram of the model construction (a) The flow diagram of stage 1; (b) The flow diagram of stage 2.

4. Strategy of selecting an access point

4.1. Principle of utilizing light-path blocking to resist interference

For ease of description, FACE AP and BACK AP respectively represent the VLC AP which the user is facing and the VLC AP which the user is facing away from. When the user communicates with the FACE AP, his body can block interference signals from the BACK AP. The anti-interference diagram is shown in Fig. 6.

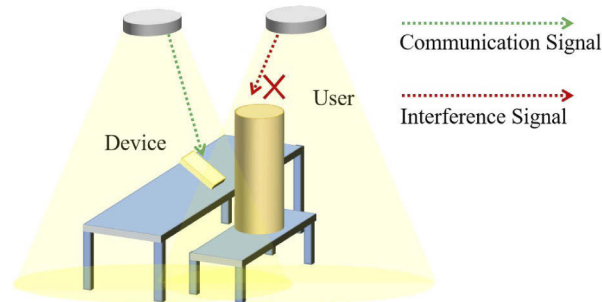


Fig. 6. The anti-interference diagram.

4.2. Implementation of the strategy

4.2.1. Judging whether the user is in the coverage range of a VLC AP

For ease of description, define d_{level} , h and R as the horizontal distance between the users' mobile devices and VLC APs, the height of the mobile device from the ground, the radius of AP's coverage at h from the ground. When $d_{levelij} < R$, it means that the user_i is in VLC AP_j's coverage range.

4.2.2. Finding FACE APs and BACK APs

Here, define d_{levelu} , $d_{levelu1}$ and $d_{levelu2}$ as the horizontal distance between (x_u, y_u) and the VLC AP's coordinate, the horizontal distance between (x_{u1}, y_{u1}) and the VLC AP's coordinate and

that between (x_{u2}, y_{u2}) and the VLC AP's coordinate respectively. As mentioned in section 3.2, judgments of user's orientations can be divided into two categories.

Category 1: The user faces a fixed device. The user's coordinate can be calculated. When $d_{levelij} < d_{leveluij}$, VLC AP_j is user_i's FACE AP. When $d_{levelij} \geq d_{leveluij}$, VLC AP_j is user_i's BACK AP.

Category 2: The user faces the center area of the crowd. The range of User's activity can be gotten. When $d_{levelij} < d_{levelu1ij}$ and $d_{levelij} < d_{levelu2ij}$, VLC AP_j is user_i's FACE AP. When $d_{levelij} \geq d_{levelu1ij}$ and $d_{levelij} \geq d_{levelu2ij}$, VLC AP_j is user_i's BACK AP. Otherwise, the user's FACE APs and BACK APs can't be found.

4.2.3. Choosing an AP

Because VLC APs in room always distribute as Lattices, the user can face two VLC APs at most. If an area is covered by two or more VLC APs, the communication signals between the user and the VLC AP chosen will be interfered by other VLC APs' signals. Here, based on the position relationship model, an AP-selection strategy will be proposed to choose the appropriate AP and block interference signals by the user's body. Three steps will be taken as follows:

Step 1: Detection about whether the user's orientation can be judged. For ease of description, define l_1 , l_2 , θ and $\theta_{\max}$ as the line formed by (x_{u1}, y_{u1}) and (x, y) , the line formed by (x_{u2}, y_{u2}) and (x, y) , the angle between l_1 and l_2 and the critical angle respectively. In section 3.2, the user's activity-range is calculated to get his orientation. But, there is a critical angle ($\theta_{\max}$). If $\theta \leq \theta_{\max}$, the user's orientation can be gotten by calculation. Namely, his orientation can be judged. And, Step 2 will be started. If $\theta > \theta_{\max}$, the user's orientation can't be gotten by calculation. Namely, his orientation can't be judged. Then, his mobile device will be directly connected to WiFi AP.

Step 2: Selection of APs. By the number of VLC APs in an area, interference intensities of signals are classified into five levels. Based on the five levels and users' orientations, APs will be selected as follows:

Level1: When the user is in an area covered by no VLC AP, the user is in a none-interference area. But, the user can't receive any signal from VLC APs. His mobile device will be connected to WiFi.

Level2: When the user is in an area covered by one VLC AP, the user is in a low-interference area. No matter where the user is oriented, his mobile device will be connected to this AP.

Level3: When the user is in an area covered by two VLC APs, the user is in a medium-interference area. If the user is only facing one of these two APs, his mobile device will be connected to the FACE AP. If the user is facing these two APs, his mobile device will be connected to these two APs together. If the user is facing away from these two APs, his mobile device will be connected to WiFi.

Level4: When the user is in an area covered by three APs, the user is in a high-interference area. If the user is only facing one AP, his mobile device will be connected to the FACE AP. If the user is facing two APs, his mobile device will be connected to these two FACE APs. If the user isn't facing any AP, his mobile device will be connected to WiFi.

Level5: When the user is in an area covered by four APs, the user is in an extreme high-interference area. If the user is facing one or zero AP, his mobile device will be connected to WiFi. If the user is facing two APs at the same time, his mobile device will be connected to these two FACE APs.

After APs are selected, Step 3 will be started.

Step 3: Verification about the judgment of the user's orientation. Sometimes, the user's orientation will be misjudged when his orientation doesn't conform to the general behavior characteristics of human. In practice, the probability of the misjudgment is small and the influence of the misjudgment is little, too. Even so, to timely deal with the misjudgment when it

happens, here, a relevant processing method will be adopted according to the number of signal interruptions and the SINR. The specific process of this method is as follows:

For ease of description, define In , $In_{\max}$, $SINR_u$ and $SINR_{\min}$ as the number of signal interruptions, the maximum number of interruptions, the SINR of the user's mobile device and the minimum threshold of SINR respectively. If the user's orientation is misjudged, In will increase and $SINR_u$ will decrease. Here, there are $In_{\max}$ and $SINR_{\min}$. If $In < In_{\max}$ and $SINR_u > SINR_{\min}$, it indicates that the judgment of the user's orientation is right. Namely, the user chooses appropriate APs. Then, his mobile device will maintain the connection described in Step 2. Otherwise, the judgment of the user's orientation is wrong. Namely, the user chooses inappropriate APs. Then, his mobile device will be connected to WiFi.

After Step 3 is finished, a new cycle will be started to choose the next user's APs.

Obviously, under the condition that the user's orientation has been already obtained, the network can automatically choose APs by the above strategy. It means that the network can self-adaptively change with the change of the user's orientation to correctly select APs. This can effectively utilize the user's body to block interference signals in order to ensure the normal communication. The flow diagram of the AP-selection strategy is shown in Fig. 7.

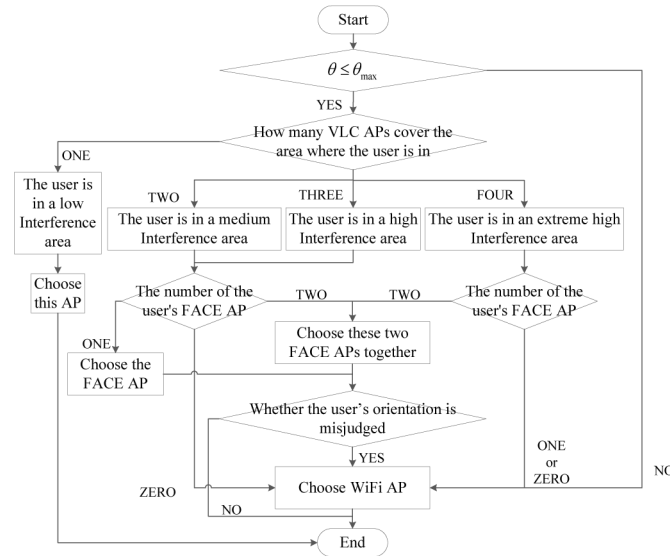


Fig. 7. The flow diagram of the strategy.

Now, it can be seen that the position relationship model has been constructed in section 3. And the AP-selecting strategy has been present in section 4. These two form a complete self-adaptable anti-interference scheme based on light-path blocking. This scheme can both determine the user's orientation by its model and self-adaptively select APs by its strategy.

In the existed schemes, firstly, device is directly connected to APs. Next, the communication quality is detected and APs will be selected by the detection results. Then, devices will be connected to the selected APs. In the proposed scheme, AP can be directly selected by the position relationship model. Thus, devices can be quickly connected to APs. Obviously, the proposed scheme has the comparative advantages: faster response speed of network, higher efficiency of user access. Besides, it is easier implementation and lower cost.

5. Simulation results

Here, the validity of the position relationship model will be simulated, the effectiveness of the AP-selection strategy will be analyzed and anti-interference performances with light-path blocking will be compared with that without light-path blocking.

5.1. Validity of the position relationship model

Here, to demonstrate the validity of the model, five categories of users' position distributions are set (shown in Fig. 8 and Fig. 9). Here, Fig. 8(a) is the graph of category1: Users face fixed devices. Figure 8(b) is the graph of category2: Users' positions distribute as a convex polygon. Figure 8(c) is the graph of category3: Users' positions distribute as a concave polygon. Figure 8(d) is the graph of category4: Multiple user-groups distribute in a scene. Figure 9 is the graph of category5: Users' positions distribute as Poisson cluster. Figure 9(a) is the enlarged diagram of users facing fixed devices. Figure 9(b) is the enlarged diagram of users facing the crowd's center.

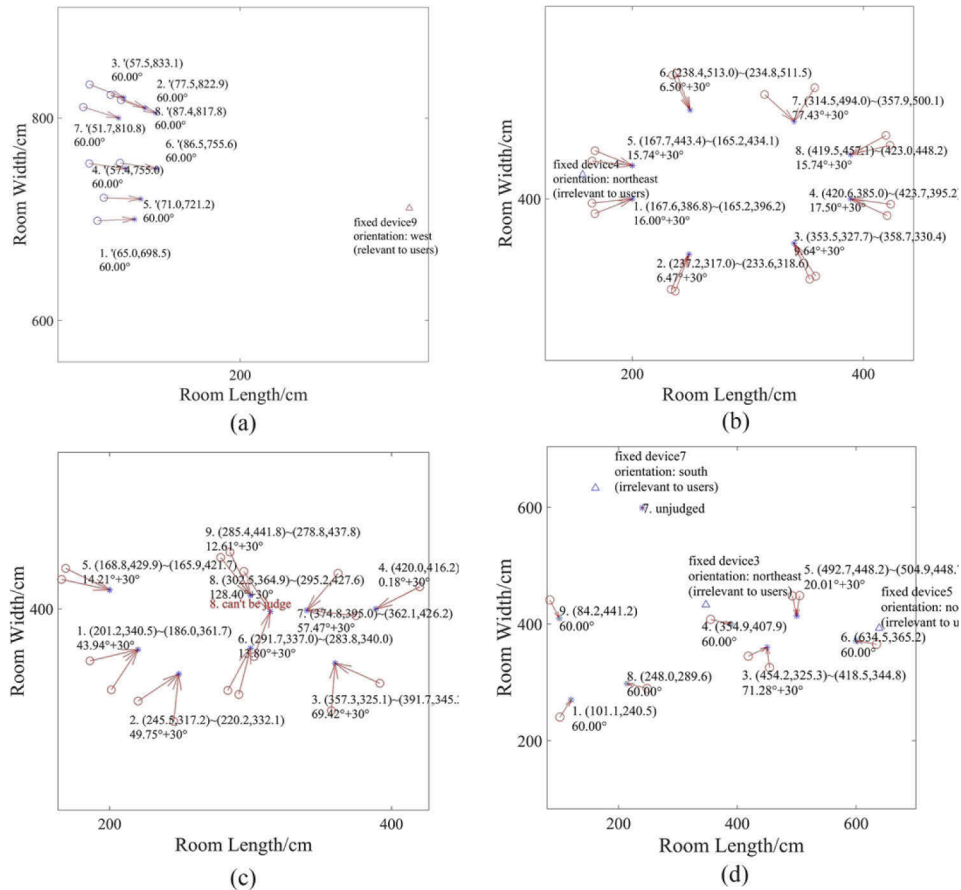


Fig. 8. The schematic diagram of users' position distributions. (a) Users face fixed devices; (b) Users' positions distribute as a convex polygon; (c) Users' positions distribute as a concave polygon; (d) Multiple user-groups distribute in a scene.

In Fig. 8 and Fig. 9, it can be seen that if only one user-group is in a scene, users' orientation can be judged correctly when users face fixed devices or users' positions distribute as a convex

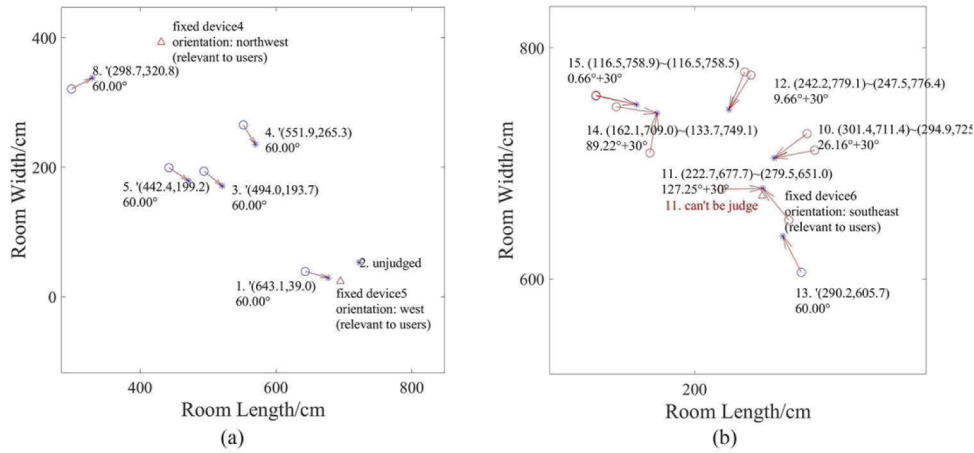


Fig. 9. Users' positions distribute as Poisson cluster. (a) The enlarged diagram of users facing fixed devices; (b) The enlarged diagram of users facing the crowd's center.

polygon. And orientations of users in a user-group's center may be misjudged when users' positions distribute as a concave polygon. It is because these users' orientations are unstable in practice. The corresponding processing method has been described in Section 4.2.

If multiple user-groups are in a scene, this model can accurately judge users' orientations when a user-group is far away from other user-groups. And this model will provide more fault tolerances for subsequent access point selection when a user-group is near other groups. The case is shown in Fig. 10.

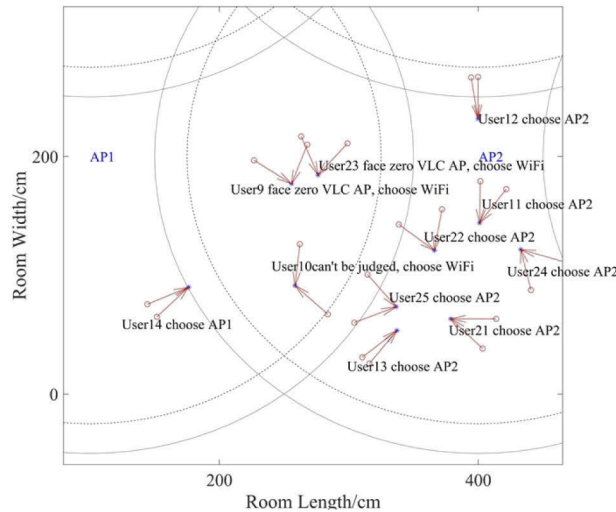


Fig. 10. The position-relationship diagram of user 10 and other users.

In Fig. 10, some users distributing as Poisson cluster are generated. To illustrate the above content, a portion of users were selected for analysis. In light of the actual situation, these users may have three relationships as follows:

Relationship 1: In Fig. 10, users are in a same user-group. Figure 10 shows that users at the edge of the group all face the center of the user-group. User22 and user25 are closer to users 11,

13, 21, and 14. So, user22 and user25 tend to face users11, 13, 21, and 24. Besides, user22 and user25 are in the low interference area, they will choose AP2 regardless of their orientation. Distances between user10 and users around him are similar. So, user10's orientation can't be correctly judged. Because he is in the middle-interference area, he will choose WiFi according to Section 4.2.

Relationship 2: Users9, 10, 14 and 23 are in Group A. Other users are in Group B. In theory, user10 should face the center of Group A. But the judging result of his activity-range is disturbed by Group B. This is because that user10 is too closer to user25.

Relationship 3: Users9, 14 and 23 are in Group A. Other users are in Group B. In theory, user10 should face the center of Group B. But the judging result of his activity-range is disturbed by Group A. This is because that user10 is too closer to user14.

In fact, these users' relationship can't be verified. With the exception of user10, judgments of other users' orientations conform to human's behavior characteristic. Because of other groups' disturbances, user10's activity-range judged by the model is larger than that in reality. So, incorrect AP-selection of user10 can be avoided, no matter which relationship between he and others is.

The above results show that this model can effectively judge the users' orientations for different user distributions.

5.2. Effectiveness of the AP-selection strategy

When one user's mobile device is connected to a FACE AP, his body can block interference signals from the BACK AP. When one user's mobile device is connected to a BACK AP, his body will block communication signals from the BACK AP. And his mobile device may receive more interference signals from the FACE AP. The farther the user is from the AP center, the higher the degree of his body's blocking the signal is. Here, define p_{rf} , p_{rb} , λ_o and λ_b as the mobile device's received optical power when the user faces the AP, the mobile device's received optical power when the user faces away from the AP, 'face coefficient' and 'back coefficient'. p_{rf} , p_{rb} , λ_o and λ_b are given by:

$$p_{rf} = \lambda_o H(0) \times p_0 \quad (7)$$

$$p_{rb} = \lambda_b H(0) \times p_0 \quad (8)$$

$$\lambda_o = 1 \quad (9)$$

$$\lambda_b = 1 - \frac{d_{level}}{R} + \delta \quad (10)$$

where $H(0)$ is the optical channel's total DC attenuation, p_0 is optical power of transmitter, δ is the compensation number approaching 0.

Here, set the inclination of the mobile device to be 20 degrees. The relational curves between the received optical power and d_{level} are shown in Fig. 11.

By comparing curves of p_{rf} and p_{rb} , it can be seen that p_{rb} is more than p_{rf} when the user is near the AP. Because the user is closer to the AP, the angle of incidence is smaller. Closer the incident light is to the vertical state, larger the received optical power will be. At the same time, the user is closer to the AP, the blocking degree of signals by the user's body is less. At this time, λ_b is close to 1. However, with the increase of d_{level} , the angle of incidence will increase and λ_b will decrease accordingly. So, p_{rb} begins to decrease. Users in the interference area are always far away from the AP. Therefore, p_{rf} is more than p_{rb} when users are in an interference area. Namely, this scheme can save more energy.

As mentioned in section 4.2, the proposed strategy is to let the user in interference area choose FACE APs. For ease of description, define $SINR_{1f}$, $SINR_{2f}$, $SINR_{1b}$, $SINR_{2b}$, $SINR_{fb}$ as the SINR of the mobile device when the user chooses one FACE AP, the SINR of the mobile device when the user chooses two FACE APs, the SINR of the mobile device when the user chooses one

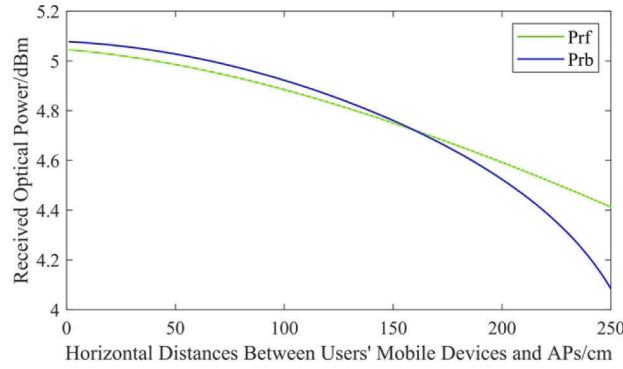


Fig. 11. The curve graph of p_r and d_{level} .

BACK AP, the SINR of the mobile device when the user chooses two BACK APs and the SINR of the mobile device when the user chooses one FACE AP and one BACK AP respectively. $SINR_f$ and $SINR_b$ represent users choosing FACE APs and users choosing BACK APs respectively.

When users are in the medium-interference area, the relational curves between SINRs and d_{level} are shown in Fig. 12(a) and Fig. 12(b). Figure 12(a) is the curve graph of SINRs when the user chooses one AP. Figure 12(b) is the curve graph of SINRs when the user chooses two APs. When users are in the high-interference area, the relational curve between the SINRs and d_{level} are shown in Fig. 12(c) and Fig. 12(d). Figure 12(c) is the curve graph of SINRs when the user chooses one AP. Figure 11(d) is the curve graph of SINRs when the user chooses two APs. When users are in the extreme high-interference area, the relational curve between the SINRs and d_{level} are shown in Fig. 12(e).

In Fig. 12(a), (c), $SINR_{1f}$ is always more than $SINR_{1b}$. It is because the user's body will block the interference signal when he chooses the FACE AP. And his body will block the communication signal when he chooses the BACK AP. In Fig. 12(b), when the user is near the AP, $SINR_{2b}$ is more than $SINR_{2f}$. When the user is far away from the AP, $SINR_{2b}$ will decrease seriously. And $SINR_{2b}$ is smaller than $SINR_{2f}$. Similarly, in Fig. 12(d), when the user is far away from the AP, $SINR_{fb}$ is smaller than $SINR_{2f}$, too. And in Fig. 12(e), $SINR_{2f}$ is always more than $SINR_{2b}$, too.

Here, take a comprehensive analysis of above situations.

If the number of communicating APs is equal to the number of interfering APs, $SINR_f$ is always more than $SINR_b$.

If the number of interfering APs is less than the number of communicating APs, $SINR_f$ is more than $SINR_b$ when users are far away from VLC APs. The further away the user is from the AP, the more $SINR_f$ is than $SINR_b$.

In a word, $SINR_f$ is always more than $SINR_b$ when users are in the interference area. It indicates that this AP-selection strategy is effective in resisting interference.

5.3. Simulations and comparisons of SINRs and WiFi network loads

Under the condition of users Poisson cluster distribution, the proposed self-adaptable anti-interference scheme is simulated. By its model, users' orientations can be gotten. By its strategy, APs can be selected. Accordingly, users' SINRs and WiFi network's loads can be also obtained. The distribution curve graph of every user's SINRs and the relation curve graph of WiFi network's loads and the number of users are shown in Fig. 12. In Fig. 12, Fig. 12(a) is the curve graph of every user's SINRs, Fig. 12(b) is the curve graph of WiFi network's loads. Here, $SINR_{blocked}$ and $SINR_{unblocked}$ respectively represent the SINR when interference signals are blocked and the SINR when interference signals are unblocked. And $LOAD_a$ and $LOAD_b$ respectively represent

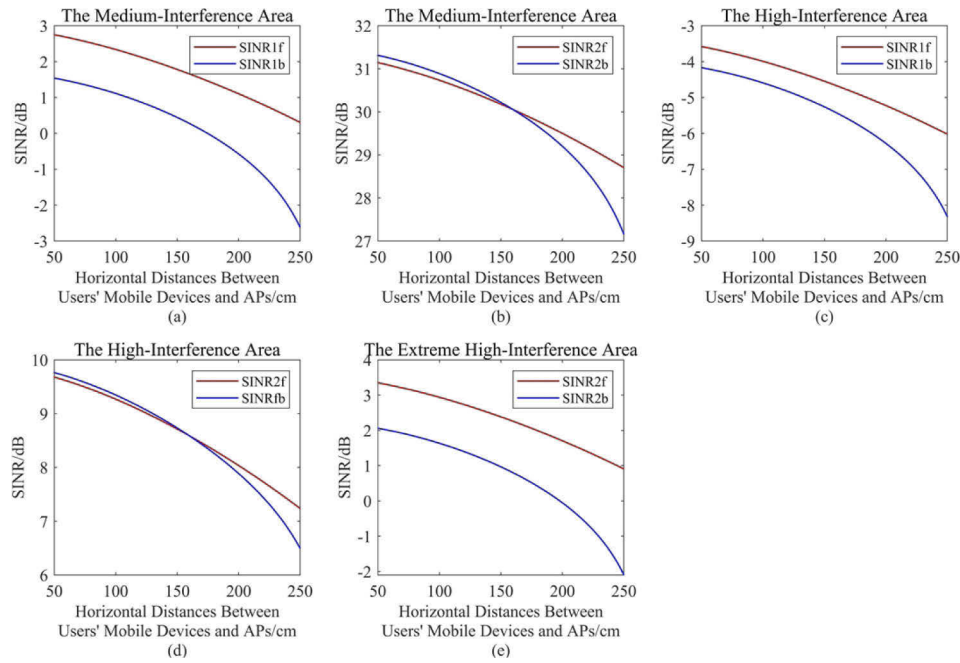


Fig. 12. The curve graph of SINRs in the interference area (a) In the medium-interference area, users choose one AP; (b) In the medium-interference area, users choose two APs; (c) In the high-interference area, users choose one AP; (d) In the high-interference area, users choose two APs; (e) In the extreme high-interference area, users choose two APs.

WiFi network's loads when the user chooses APs by the proposed scheme and WiFi network's loads when any interfered user chooses WiFi APs.

In Fig. 13(a), the magenta curve and the blue curve respectively represent curves of $\text{SINR}_{\text{blocked}}$ and $\text{SINR}_{\text{unblocked}}$. Comparing these two curves, it can be seen that $\text{SINR}_{\text{blocked}}$ is always more than $\text{SINR}_{\text{unblocked}}$ if the user is in the interference area. This indicates that the self-adaptable anti-interference scheme can resist interference and increase SINR effectively.

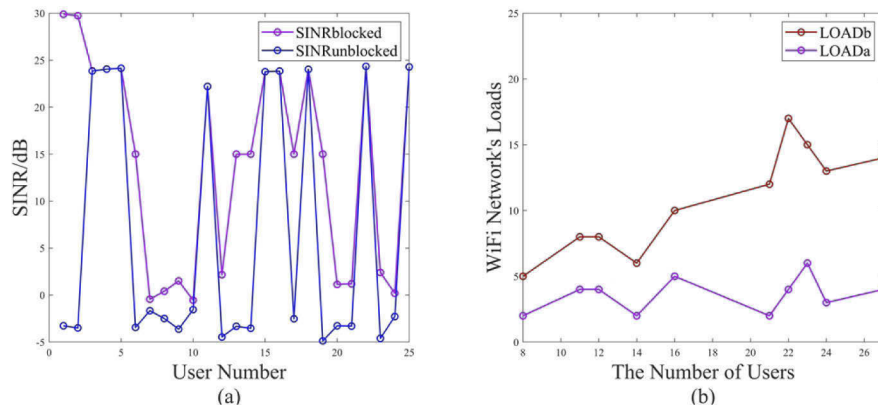


Fig. 13. The curves of every user's SINRs and that of WiFi network's loads (a) Every user's SINRs; (b) WiFi network's loads.

In Fig. 13(b), the magenta curve and the red curve respectively represent curves of $LOAD_a$ and $LOAD_b$. Comparing these two curves, it can be seen that $LOAD_a$ is always smaller than $LOAD_b$. As the number of users increases, the difference between $LOAD_a$ and $LOAD_b$ becomes more pronounced. This indicates that the proposed scheme can still decrease the load of WiFi in the hybrid WiFi-VLC network. It means that the scheme can effectively lift availability of the network.

6. Conclusion

Now, the self-adaptable anti-interference scheme by light-path blocking in hybrid WiFi-VLC network has been finished. At first, a user-device position relationship model was constructed by human behavior characteristics. In the model, users' orientations can be determined according to different distributions of users. Then, based on the model, a strategy of users' choosing APs in different interference areas was present. Next, based on the strategy, communicating APs can be self-adaptively selected to match them with users' orientations. In the scheme, interference signals can be effectively blocked by users' bodies to ensure normal communication. Finally, simulation results indicated that this scheme can effectively resist interferences between VLC APs by light-path blocking. Compared with other schemes, the scheme not only can save energy, increase SINR and lift availability of the hybrid WiFi-VLC network, but also has faster response speed of network, higher efficiency of user access, easier implementation and lower cost.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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